

CLINTON, OK, USA

WMO: 720356

Lat: 35.538N

Lon: 98.933W

Elev: 493

StdP: 95.55

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 13999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.5	-18.0	0.8	-2.7	-16.2	1.0	-0.9	13.6	8.8	12.4	7.7	4.1	0	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	38.7	19.1	37.3	20.1	35.9	20.6	25.2	31.8	24.2	30.8	23.3	29.1	5.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.7	27.8	22.5	18.2	26.6	22.0	17.7	26.1	80.5	31.1	75.9	31.3	72.1	28.4	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.3	11.1	9.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	3.5	4.7	10.9	15.1	19.9	25.6	27.7	27.5	22.8	15.8	8.9	3.8
	DBStd	10.00	5.20	6.08	5.78	5.12	4.72	3.27	2.91	3.24	3.91	5.42	5.08	4.97
	HDD10.0	742	209	166	59	14	1	0	0	0	0	14	79	199
	HDD18.3	2094	460	382	236	121	37	0	0	0	8	113	286	451
	CDD10.0	2777	8	17	88	167	309	467	548	543	384	194	46	7
	CDD18.3	1088	0	0	6	23	87	217	289	285	143	35	2	0
	CDH23.3	12723	1	15	76	283	953	2517	3615	3418	1454	353	36	1
	CDH26.7	6484	0	3	14	85	393	1292	2025	1912	655	98	7	0
Wind	WSAvg	4.4	4.3	4.6	5.3	5.4	4.9	4.8	3.9	3.5	3.8	4.3	4.5	4.1
Precipitation	PrecAvg	787	27	30	61	75	104	105	61	96	68	75	44	37
	PrecMax	1245	66	95	157	212	218	282	130	181	185	200	155	106
	PrecMin	496	0	0	0	4	22	8	6	0	1	5	0	4
	PrecStd	169	19	28	46	45	53	67	32	50	48	52	45	30
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.4	25.3	28.0	32.4	35.8	40.1	40.9	41.4	37.3	31.3	27.2	20.5
		MCWB	10.6	11.8	13.7	17.6	18.1	18.7	18.6	18.5	18.3	19.6	14.7	10.5
	2%	DB	17.9	21.2	25.2	28.9	32.9	37.3	37.9	39.0	34.2	29.8	22.9	17.3
		MCWB	8.1	10.9	13.8	16.8	18.6	19.2	19.8	19.4	19.8	19.0	13.4	10.1
	5%	DB	15.9	17.6	22.6	26.4	31.2	35.1	37.2	37.3	32.5	27.4	21.0	14.9
		MCWB	7.3	9.3	13.8	15.7	18.4	20.2	20.3	20.7	19.3	17.3	12.8	8.2
	10%	DB	12.6	15.0	20.2	24.0	28.0	32.7	35.3	35.2	31.0	25.1	17.5	12.2
		MCWB	6.1	8.3	12.7	15.2	17.9	20.1	20.9	21.4	19.3	16.7	11.0	7.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	14.7	19.2	20.4	23.5	24.8	25.4	27.6	24.1	23.1	17.7	15.3
		MCDB	17.7	17.8	22.7	25.8	29.2	30.0	32.6	33.1	29.2	28.8	23.7	17.2
	2%	WB	9.9	12.7	17.5	19.1	22.1	23.7	24.5	26.2	22.7	21.2	15.5	11.7
		MCDB	16.3	19.3	20.7	24.9	28.2	30.1	31.9	31.4	28.3	25.8	19.5	13.9
	5%	WB	7.8	10.8	16.0	17.9	20.9	22.7	23.7	25.0	21.9	19.9	13.9	9.2
		MCDB	14.3	17.0	19.5	23.5	26.2	29.1	30.5	31.2	27.4	24.8	18.8	13.8
	10%	WB	6.4	8.7	14.1	16.8	19.9	21.8	22.8	23.7	21.1	18.3	12.1	7.1
		MCDB	12.2	14.9	18.8	21.7	26.0	28.4	29.3	29.2	26.6	23.0	16.7	11.6
Mean Daily Temperature Range	5% DB	MDBR	13.1	12.9	13.4	13.3	13.0	12.8	12.9	12.8	12.6	13.7	13.8	11.6
		MCDBR	17.9	18.3	16.3	16.2	15.5	15.0	14.6	15.4	14.9	15.1	17.6	16.2
		MCWBR	9.8	10.0	8.4	7.5	5.5	3.9	3.3	3.5	4.4	6.7	9.4	9.5
	5% WB	MCDBR	16.2	16.8	12.1	13.6	12.2	12.2	12.3	12.3	11.5	11.3	14.7	13.9
		MCWBR	9.6	10.1	7.4	7.7	5.8	5.5	4.8	5.0	5.0	6.1	9.6	9.6
Clear-Sky Solar Irradiance	taub	0.281	0.291	0.332	0.366	0.389	0.414	0.413	0.411	0.375	0.348	0.307	0.284	
	taud	2.527	2.508	2.408	2.343	2.350	2.323	2.346	2.305	2.382	2.440	2.494	2.518	
	Ebn at Noon	932	959	938	917	896	870	867	863	881	878	889	905	
	Edh at Noon	79	91	110	124	125	128	125	127	112	97	82	76	
All-Sky Solar Radiation	RadAvg	2.93	3.57	4.71	5.82	6.32	7.03	7.00	6.32	5.30	3.99	3.06	2.48	
	RadStd	0.27	0.47	0.32	0.44	0.65	0.59	0.38	0.37	0.35	0.52	0.39	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page